

SportSlotReservation

Phase 3 Engineering Report — Booking Engine

Field	Value
Scope	Facilities, policies, computed availability, distributed-lock booking, cancellation, audit trail
Period	2026-06-12 (sub-phases 3.1–3.6, incl. corrective 3.6.1)
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Status	PHASE 3 COMPLETE — every guard validated live; concurrency proof passed
Tests / coverage	75 passing · 90% (gate 80%)
New infrastructure	Memorystore Redis (Basic 1GB, asia-south1, AUTH via Secret Manager), Direct VPC egress
ADRs	0009 Slot Locking (Redis) · 0010 Booking Domain & Policy · 0011 Audit Logging

1. Executive Summary

Phase 3 delivered the complete booking lifecycle for the multi-tenant platform: tenant-admin-managed facilities, a policy engine resolving tenant overrides over global defaults, availability computed on demand (no slot documents are ever pre-generated), booking creation guarded by a Redis distributed lock plus a transactional quota-and-duplicate check, buffer-enforced cancellation with actor attribution, and a synchronous Firestore audit trail. A 20-way concurrency test proved exactly-one-winner semantics; stopping Redis live demonstrated fail-closed behavior (503, never a lock bypass).

All wall-clock policy rules (the 20:00 booking window, day-quota boundaries, cancellation deadlines) evaluate in the **tenant's timezone** while all stored timestamps remain UTC — validated live with the server on UTC and rules tracking IST.

2. Key Decisions

Decision	Choice	Rationale
Slot locking	Memorystore Redis from day one (SET NX PX 10s, owner-checked Lua release)	Production parity now, funded by trial credits; Firestore-only locking degrades under flash traffic
Lock failure	Redis down → 503 LOCK_UNAVAILABLE; held → 409 SLOT_CONTENTENDED	Charter Fail Closed: never bypass the lock
Availability	Computed: bookings are the only stored slot artifacts	No generation jobs; avoids ~5.8M empty documents/year at target scale
Booking identity	Deterministic ID {facility}_{date}_{start}	Natural idempotency; the ID doubles as the lock key and the duplicate guard
Quota	Checked inside the Firestore transaction	The lock guards a slot; quota races across slots — pre-checks are racy
Cancellation	Self or tenant_admin only; buffer on tenant clock; attribution recorded	Household-member cancels invite disputes; admin acts are auditable
Audit	Synchronous Firestore events /tenants/{id}/audit (ADR-0011)	BigQuery banned from request path (200–300 ms streaming inserts; Cloud Run post-response CPU throttling)
In-progress slots	Bookable for the remainder, marked IN_PROGRESS + response notice	Mark-don't-hide; the UI warns before confirming (Phase 4)

3. The Booking Gauntlet

A booking request passes, in order: schema validation → facility existence/active → server-side recomputation of the slot's bookability (identical code path to the availability display, so UI and enforcement cannot drift) → Redis lock acquisition → Firestore transaction (same-day confirmed count vs. quota; status-aware duplicate check) → audit event → lock release. The 20-way concurrency run showed the layers in action: 1×201 winner, late arrivals stopped by the cheap pre-check (422 BOOKED), racers stopped by the lock (409 SLOT_CONTENTENDED), and same-user attempts stopped by quota — with a rerun accidentally proving the rate limiter as well (exactly 10 of 30-per-minute over-budget requests received 429).

4. Issue Log

#	Issue	Rule adopted
1	Worker reports proposed off-roadmap next steps	Worker suggestions are input, never plan
2	In-progress slots silently bookable (emergent, undecided)	Emergent behaviors found in validation get an explicit ruling
3	Silent curls (server down) could read as passes	Validation curls always print the status code
4	Audit/BigQuery decision existed only in conversation	Every decision lands in an ADR or it does not exist

#	Issue	Rule adopted
5	Cancelled slots advertised bookable but rejected all rebooking (found by deployed probe, not the 75-test suite)	Trace every lifecycle loop (create→cancel→recreate) in close-out review

5. Live Validation Record

Probe	Result
Role gate: resident creates facility	403 FORBIDDEN_ROLE
tenant_admin creates facility	201 (first tenant-admin write)
Availability: tenant-clock past/window/horizon	PAST · WINDOW_NOT_OPEN · BEYOND_HORIZON (IST, server UTC)
First booking	201 — Redis lock + transaction + real Firestore document
Duplicate / quota / contention	422 BOOKED · 409 QUOTA · 409 CONTENTED
Redis stopped, booking attempted	503 LOCK_UNAVAILABLE — fail closed, no bypass
Cancellation matrix	200 self · 403 other-resident · 422 too-late · 409 already-cancelled
Concurrency (20 parallel, one slot)	Exactly one 201 — PASS
Rebooking a cancelled slot (post-3.6.1)	201 by a different user — lifecycle closed

6. Phase 4 Preview

Frontend foundation: React 18 + Vite + TypeScript PWA on Firebase Hosting, wildcard-subdomain tenant resolution and DNS (which finally exercises the tenant cross-check over the public edge), Firebase Auth UI flows, the booking screens against the Phase 3 API — including the IN_PROGRESS warning dialog — and per-tenant branding via CSS variables.